

Java Practical Questions – Inheritance, Interfaces, Packages & Exception Handling

1. Method Overriding

Write a Java program to create a superclass Animal with a method sound(). Create subclasses Dog and Cat that override the sound() method. Display the appropriate sound for each animal.

2. Dynamic Method Dispatch

Write a Java program to demonstrate **dynamic method dispatch** using a superclass Shape and subclasses Circle and Rectangle. Override the draw() method in each subclass and invoke it using a superclass reference.

3. Abstract Class

Write a Java program to create an abstract class Vehicle containing an abstract method start() and a concrete method display(). Create subclasses Car and Bike and implement the start() method in each subclass.

4. Defining and Implementing an Interface

Define an interface Printable with a method print(). Create classes Student and Teacher that implement the interface. Write a program to display the details of a student and a teacher using the print() method.

5. Multiple Interfaces

Write a Java program to demonstrate the implementation of **multiple interfaces**. Define interfaces Sports and Academics, each containing one method. Create a class Student that implements both interfaces and displays the student's academic and sports information.

6. Packages – Declaring and Importing a Package

Create a package named college containing a class Student with a method to display student details. Write another Java program outside the package to **import the college package** and access the Student class.

7. Sub-packages

Create a package structure college.department. Define a class ITStudent inside the department sub-package with a method to display student information. Write another Java program to import and use this class.

8. try-catch Exception Handling

Write a Java program to accept two integers from the user and perform division. Use **try-catch** to handle the ArithmeticException that occurs when the denominator is zero.

9. throw and throws

Write a Java program to create a method checkAge(int age) that throws an exception using the **throw** keyword when the age is less than 18. Declare the exception using **throws** and handle it in the calling method.

10. Multiple Exceptions and finally

Write a Java program that accepts an array index and performs an operation on an array. Use try-catch to handle ArrayIndexOutOfBoundsException and another appropriate exception. Use a **finally** block to display a message indicating that exception handling has been completed.